SE Lab 1

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5-J

Vaccination Cohort & Dose Scheduling System

REQUIREMENTS TABLE

REQ ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE
FR-001	Functional	The system shall enforce minimum dose interval rules before allowing a citizen to book the next vaccination dose.	HIGH	**Pass:** Dose 2 booking is blocked when fewer than 28 days have elapsed since Dose 1. **Fail:** The system allows an early booking.
FR-002	Functional	The system shall allow a citizen to register for a vaccination cohort by providing the required demographic information.	HIGH	**Pass:** A citizen providing all required information receives a unique registration record. **Fail:** Registration is completed with missing mandatory information.
FR-003	Functional	The system shall allow a citizen to select and book an available vaccination centre slot for an eligible dose.	MEDIUM	**Pass:** A valid available slot is reserved and shown in the citizen's booking record. **Fail:** The system confirms a slot that is unavailable or already fully booked.
FR-004	Functional	The system shall allow a vaccination officer to record the administration of a vaccine dose for a registered citizen.	MEDIUM	**Pass:** After successful recording, the citizen's vaccination history contains the administered dose and date. **Fail:** The dose is not reflected in the vaccination record.
FR-005	Functional	The system shall generate a digitally verifiable QR vaccination certificate after a citizen's vaccination record is successfully updated.	LOW	**Pass:** A QR certificate is generated and can be successfully verified using the system. **Fail:** A certificate is generated but its QR code cannot be verified.
NFR-001	Non Functional	The vaccination certificate verification endpoint shall authenticate digitally signed QR codes offline/online in under **150 ms**.	HIGH	**Pass:** Benchmarking confirms verification completes in under 150 ms under simulated peak load.

				Fail: Verification exceeds 150 ms or accepts an invalid signature.
NFR-002	Non Functional	The system shall prevent double-booking of a vaccination centre slot.	HIGH	**Pass:** When a slot reaches its capacity, subsequent booking attempts are rejected. **Fail:** Two citizens are confirmed for the same remaining slot capacity.

Rationale

FR-001: Prevents citizens from receiving subsequent doses before the required interval.

FR-002: Demographic registration is required to organize vaccination rollouts.

FR-003: Enables organized vaccination scheduling and prevents overbooking.

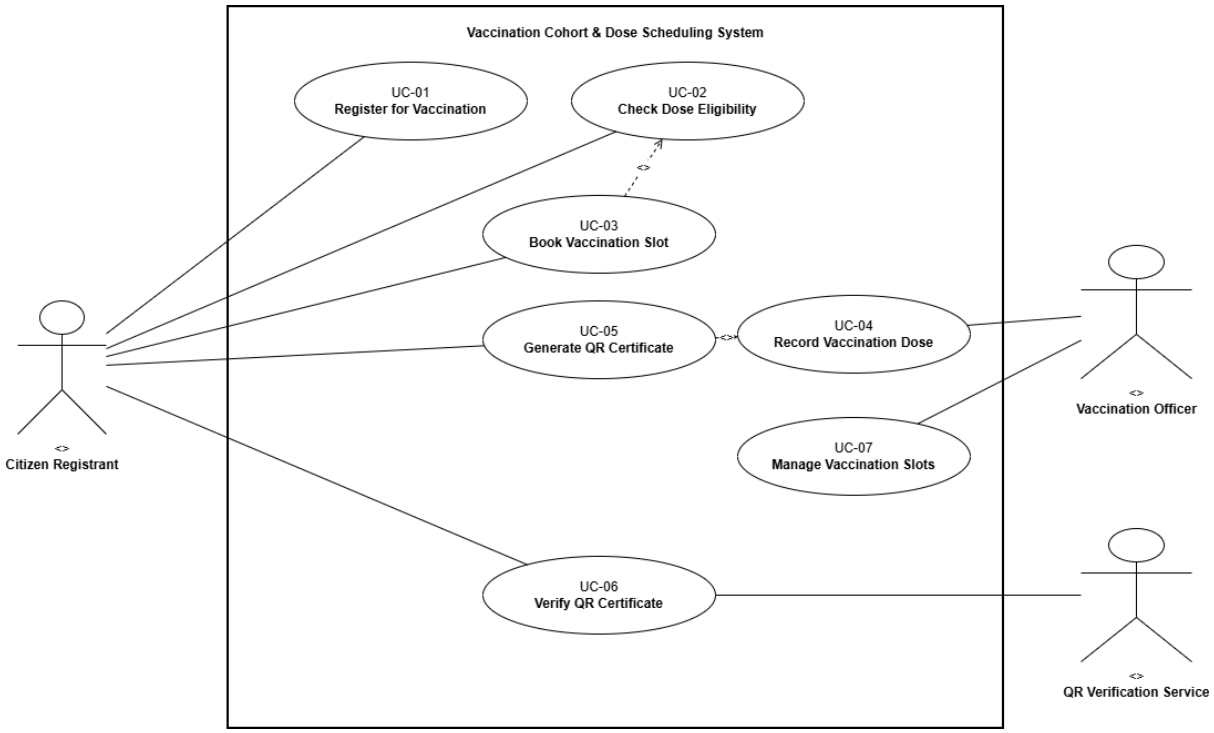
FR-004: Accurate dose records are required for enforcing future dose intervals and certificates.

FR-005: Provides citizens with a verifiable digital record of vaccination.

NFR-001: Enables fast and secure verification of vaccination certificates.

NFR-002: Prevents scheduling conflicts and ensures reliable centre capacity management.

UML DIAGRAM



Use Case: Book Vaccination Slot

1. Preconditions

1. The citizen has successfully registered in the vaccination system.
2. The citizen has provided the required demographic information.
3. The citizen is eligible for the requested vaccination dose.
4. At least one vaccination centre has an available slot.

2. Postconditions

Success:

- The selected vaccination slot is reserved for the citizen.
- The booking details are stored in the system.
- The citizen can view the scheduled vaccination appointment.

Failure:

- No booking is created.
- The citizen is informed that the requested slot cannot be booked.

3. Main Success Scenario

1. The **Citizen Registrant** logs into the vaccination system.
2. The citizen selects **Book Vaccination Slot**.
3. The system displays the citizen's vaccination history and requested dose.
4. The system checks the minimum required interval since the previous dose.
5. The system confirms that the citizen is eligible for the requested dose.
6. The system displays available vaccination centres and their available slots.
7. The citizen selects a preferred vaccination centre and time slot.
8. The system checks the availability of the selected slot.
9. The system reserves the slot for the citizen.
10. The system displays the booking confirmation.
11. The use case ends successfully.

4. Alternate Flow — Minimum Dose Interval Not Met

4a. Dose interval requirement not satisfied

1. At Step 5, the system determines that the minimum required interval since the previous dose has not elapsed.
2. The system blocks the vaccination booking.
3. The system displays the earliest date on which the citizen can book the next dose.

4. The citizen is returned to the vaccination scheduling screen.
5. The use case ends without creating a booking.